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How GPT-5 helped immunologist Derya Unutmaz solve a 3-year-old mystery

The model's ability to augment human expertise could help advance fields including cancer research, autoimmune disease, and infections.

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Doctor and immunologist Derya Unutmaz has been interested in artificial intelligence for years. But his “aha” moment came in late 2025, when GPT-5 Pro helped him and his lab revisit a three-year-old puzzle centered on a special type of immune cell that helps the human body fight cancer and other illnesses.

The mystery centered on a basic but consequential question in immunology: how does glucose affect the way T cells develop and specialize? T cells are immune cells that help the body fight viruses, kill cancerous cells, respond to some bacteria and parasites, and distinguish healthy cells from threats. As they develop, they take on different jobs, including roles that can shape cancer, autoimmune disease, and infection.

Understanding what pushes T cells toward one specialization or another could help researchers better understand, and eventually, better treat those diseases.

Today, Unutmaz—a professor at The Jackson Laboratory and the University of Connecticut—says AI has become so central to his work that he can't imagine doing science without it. “That would be like taking both of your hands away, or half of your brain away,” Unutmaz said.

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understand how a type of sugar called glucose affected the development of T cells. The cells use glucose as a fuel source, but also to build proteins and carry out other functions.

The results of Unutmaz's experiment could have implications for ailments like cancer, autoimmune disease, and infections. But at the time, Unutmaz and his lab couldn't make sense of what they were seeing.

Solving a problem with GPT-5 Pro

Previous studies provided strong evidence that glucose metabolism influenced how T cells specialize. To better understand this relationship, Unutmaz and his team exposed T cells early in their development to either a low-glucose environment or to one containing a glucose-like molecule called deoxyglucose. Deoxyglucose interferes with a cell's ability to use glucose, disrupting energy production and protein construction. Proteins matter because they coordinate activity within a cell and act as messengers that send and receive information outside the cell.

The team expected the two conditions to produce similar results. In both cases, glucose, and therefore the energy the T cells needed to function, would be limited. But that's not what happened.

The T cells exposed to deoxyglucose overwhelmingly produced cells involved in the body's inflammatory response. Some of the T cells exposed to low concentrations of glucose specialized as inflammatory-response cells, but not at the numbers seen for deoxyglucose. The effects of early exposure to deoxyglucose persisted even when researchers removed the glucose-like molecule.

This difference couldn't be attributed to a lack of energy alone. Something else was going on. But Unutmaz and his lab were unable to figure out what was happening, so they shelved the experiment and moved on to other urgent tasks that needed their attention.

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experiment. He uploaded the results into the model and asked it to analyze the data.

GPT-5 Pro suggested that deoxyglucose interfered with the construction of a protein called IL-2. This protein can prevent T cells from becoming an inflammatory-response cell known as Th17. Deoxyglucose essentially removed a barrier to a T cell's ability to become a Th17 cell. That's potentially why T cells in the low-glucose environment didn't become Th17 cells at nearly the numbers seen in the deoxyglucose environment.

"GPT-5 came up with this really remarkable insight that retrospectively, makes perfect sense," Unutmaz said. It was just enough outside of his own area of expertise that he didn't see the connection himself, and neither did anyone in his lab.

Unutmaz then decided to see if GPT-5 could predict the outcome of an experiment. The immunologist started with one he had already conducted on a T cell that targets a type of lymphoma. His experiment showed that these particular T cells, called CD8+, had an enhanced ability to kill the lymphoma cells.

When Unutmaz asked GPT-5 Pro to simulate the same experiment, it correctly predicted the boost in the CD8+ cells' ability to kill lymphoma cells. The model couldn't have gleaned the results from the internet because Unutmaz hadn't yet published the results.

"That was the moment that I felt like, okay, these models have now come to a point where they really, truly understand," he said.

What this means for scientific research

Unutmaz said that models like GPT-5 Pro function more like collaborators now. They can streamline literature reviews, processing hundreds of new academic papers published every week and helping scientists identify questions that remain unanswered. They can also help researchers hone their hypotheses, reducing the amount of time it takes to identify the most worthwhile experiments to conduct.

"The number of things you can do to address your hypothesis is vast," Unutmaz said.

"You have countless approaches, and you don't know which one will be the best strategy." So he uses GPT-5 Pro to simulate experiments and predict outcomes to

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of biology.

Despite this, subject matter expertise is still key. AI may generate an insight, but people must still evaluate its significance and plausibility. For instance, someone without Unutmaz's expertise wouldn't have been able to tell if the mechanistic insight GPT-5 Pro flagged in his immune cell experiments was important or not.

The ability to generate insights and accelerate work is why these capabilities need to be handled responsibly. AI could help researchers move faster in biology and medicine, but those capabilities could also lower barriers for misuse, including by bad actors seeking to design or use biological or chemical weapons. OpenAI's Preparedness Framework outlines our approach to tracking these risks and building safeguards against AI capabilities that could create severe harm.

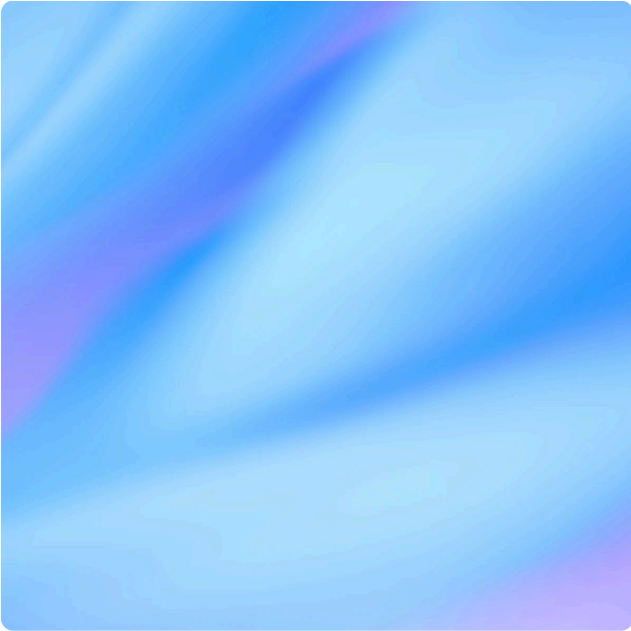
Unutmaz is optimistic about where AI is headed. It's unlike anything that has come before, he says—not the internet or the industrial revolution. Most recently, Unutmaz has experimented with advanced AI tools, including Codex and GPT-5.2 Deep Research, to help compile large-scale cancer mutation datasets and generate research materials—including an extensive T-cell-focused draft textbook—aimed at accelerating efforts in precision immunotherapy.

Unutmaz feels fortunate to be part of this time of discovery. “To not only be able to witness it historically but participate a little bit, I feel truly lucky and privileged to do that.”

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